

Mean-Shift tracking

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I. INTRODUCTION

In this assignment, the mean shift method was implemented and used to develop a tracker, which was evaluated on the VOT14 dataset. The model's performance was analyzed under different parameter settings. The tracker was further enhanced by incorporating additional weights into the histogram, with the weights adjusted based on the background characteristics of the tracked object.

II. EXPERIMENTS

A. Mean shift mode seeking on the given example

The Mean Shift method was tested with various parameters to evaluate its performance. The kernel size, which defines the region for the calculation of the current step, was adjusted. Larger kernels led to faster convergence, but excessively large or small kernels prevented convergence to the local maximum. Different starting positions were tested, showing that starting points in low-probability areas prevented movement, and for multi-modal functions, the starting point determined which maximum the algorithm would reach.

Three convergence criteria were evaluated: (1) step size, stopping when movement is under one pixel; (2) Euclidean distance, halting when consecutive positions differ by less than two units; and (3) change in probability, stopping when the probability shift is minimal. The sub-pixel step size was the most consistent but slowest, while the probability-based criterion sometimes caused premature convergence if early changes were small. Figure 1 shows how these parameters influenced convergence. Note that each color represents a different starting position.

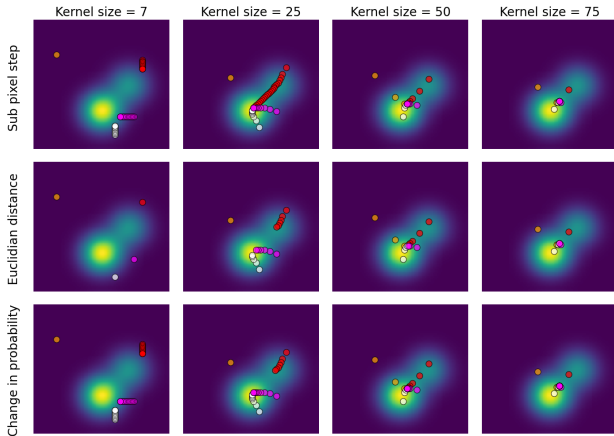


Figure 1. Comparison of the mean shift method with different kernel sizes, convergence criteria and starting positions

B. Mean shift mode seeking on custom examples

The algorithm was then tested on three custom functions, shown in Figure 2. In these tests, the sub-pixel convergence criterion was used, and the kernel size was set to 25. For each function, four different starting positions were evaluated. The

first function is a Gaussian distribution, where the algorithm consistently converges to the global maximum, provided the starting point is not in a region with near-zero probability. The second example is the Laplacian of Gaussian (the Mexican hat function), which has multiple local maxima. Here, the convergence outcome strongly depends on the starting position. The final example is a small section of the Julian Alps (around 46° North and 14° East). Due to the presence of many local maxima, the starting position determines which peak the algorithm will ascend to.

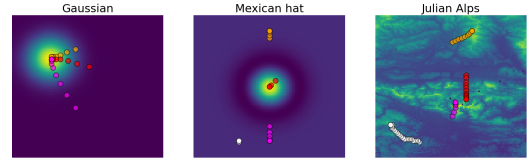


Figure 2. Comparison of the mean shift method on different functions

C. Basic tracker implementation

The tracker was implemented and tested on the entire VOT 2014 dataset. The number of failures and the tracking speed (measured on my personal laptop) are summarized in Table I. Overall, the tracker failed a total of 33 times across the dataset. While the tracker performs reliably in most cases, it struggles more with certain sequences, such as hand2, fish1, torus, and tunnel. The reasons behind these failures will be explored in detail in the next section.

Table I
BASIC TRACKER RESULTS

Sequence	Failures	Speed	Sequence	Failures	Speed
ball	1	1823 FPS	david	1	1042 FPS
basketball	0	510 FPS	diving	0	1328 FPS
bicycle	1	1047 FPS	drunk	1	700 FPS
bolt	2	1425 FPS	fernando	2	358 FPS
car	0	2017 FPS	fish1	3	1571 FPS
fish2	1	1224 FPS	motocross	2	504 FPS
gymnastics	0	1667 FPS	polarbear	0	2020 FPS
hand1	2	923 FPS	skating	1	1098 FPS
hand2	5	2610 FPS	sphere	0	1285 FPS
jogging	1	1704 FPS	sunshade	0	928 FPS
surfing	0	3179 FPS	torus	3	1170 FPS
trellis	2	1449 FPS	tunnel	4	2684 FPS
woman	1	2525 FPS			

D. Failure cases discussion

Figure 3 shows the cases where the tracker experienced the most failures. In the fish1 and hand2 sequences, failures occurred due to similar color distributions between the tracked object and the background, making it hard to detect movement. In the torus and tunnel sequences, scale changes added complexity: in torus, object rotation altered the color distribution, and in tunnel, the motorbike's decreasing size led to an inaccurate bounding box.

These failure cases suggest two potential improvements: addressing scale changes by testing different template scales and minimizing the influence of the background. The latter is discussed in the final section.

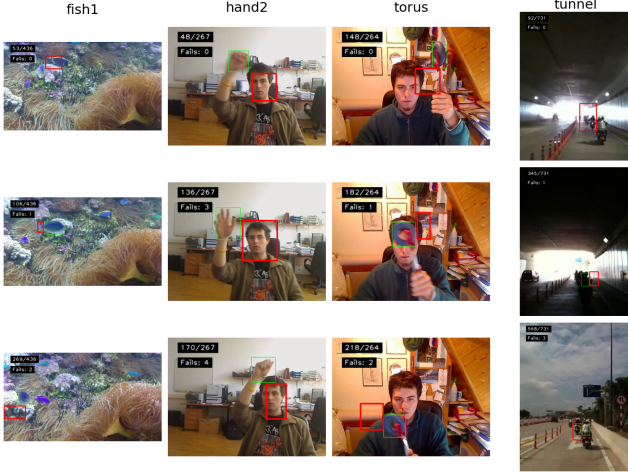


Figure 3. Mean shift tracker failure cases

E. Parameter tuning

Four key parameters were tested: σ (kernel size), histogram bin count, iterations per frame, and α (template update rate). The algorithm was evaluated for all combinations of 4 bin counts, 5 kernel sizes, 3 iteration values, and 6 update rates.

To determine the optimal number of iterations and bins, the results were averaged with respect to these two parameters (Figure 4). The data shows that using only one iteration significantly reduces performance, while 10 and 20 iterations produce similar results. Since the average performance is very similar between 10 and 20 iterations, the 20-iteration version will be used for further evaluation, as it produced the top 3 best single results, including the best result with 25 failures across the entire dataset.

Regarding the number of bins, 8 and 16 bins outperform the others and deliver a really similar performance. However, since the top 7 single results were produced by the 16-bin version, this version will be used for further evaluation.

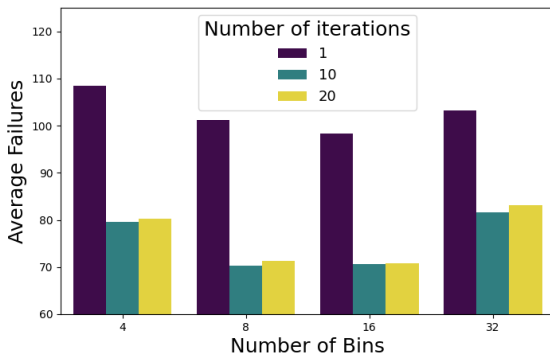


Figure 4. Performance of the means shift tracker with different numbers of bins and numbers of iterations

To determine the best values for σ and α , we refer to Figure 5. The optimal σ could be 0.5 or 1, but we choose 1 since it produced the top 2 single results. For α , performance is similar across the range from 0 to 0.01, but 0.001 appears to slightly outperform the others. We can interpret that, while updating the template can help, it should be performed really slowly.

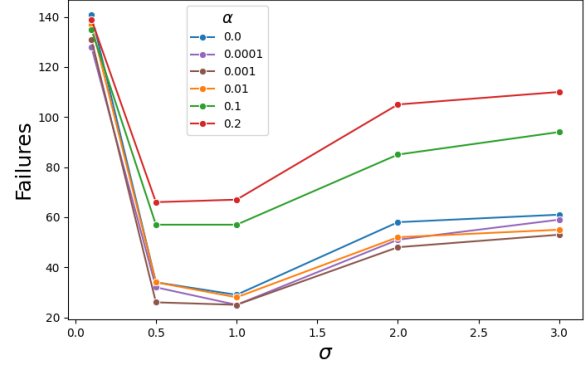


Figure 5. Performance of the means shift tracker with different kernel sizes and update rate parameters

F. Feature selection by accounting for the background in different color spaces

To mitigate the influence of the background, the method applied smaller weights to the regions in the histogram where the background color predominated. The tracker was also evaluated using five different color spaces. Additionally, a new parameter was introduced: the size of the background area. In the results, the optimal size of the background area for each color space will be taken for evaluation.

Table II presents the number of failures across the entire dataset for each color space using the optimal parameters, showing an improvement over the basic method (except for the YCrCb color space). Note that by using the HSV color space, which performed best on the entire dataset, the performance on the most challenging videos (hand2, fish1, torus, and tunnel) improved, with the number of failures reduced to 4, 1, 0, and 3, respectively.

Table II
PERFORMANCE OF THE IMPROVED TRACKER ACROSS DIFFERENT COLOR SPACES.

Color space	Failures	Average Speed [FPS]
RGB	19	981
LAB	21	831
HSV	16	1002
YCrCb	26	1230
BGR	19	1646

III. CONCLUSION

In this assignment, the mean shift algorithm and tracker were implemented. The analysis identified the problematic test sequences and explored the potential reasons for failure. The results demonstrated that the tracker's performance can be significantly improved by carefully tuning the parameters and incorporating background information into the model.